

ITA0608-MACHINE LEARNING

LAB PROGRAMS

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EXP NO:1 FIND S-Algorithm

AIM:

To implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis from the given training data

PROCEDURE:

1. Define the training dataset
2. Initialize the hypothesis with the most specific values (0).
3. Consider only the positive training examples
4. Update the hypothesis by replacing mismatched attributes with ?
5. Display the final hypothesis

PROGRAM:

```
1.import pandas as pd
```

```
data = pd.DataFrame({  
  
    'Outlook':['Sunny','Sunny','Rain  
y','Sunny','Overcast','Rainy'],  
  
    'Temperature':['Hot','Hot','Mild',  
    'Cool','Hot','Cool'],  
  
    'Humidity':['High','High','High','  
Normal','Normal','Normal'],  
  
    'Wind':['Weak','Strong','Weak','  
Weak','Strong','Strong'],  
  
    'Play':['Yes','Yes','No','Yes','Yes','
```

```
No']
```

```
}}
```

```
concepts = data.iloc[:, :-1].values
```

```
target = data.iloc[:, -1].values
```

```
hypothesis = None
```

```
for i in range(len(target)):
```

```
    if target[i] == "Yes":
```

```
        if hypothesis is None:
```

```
            hypothesis =
```

```
concepts[i].copy()
```

```
        else:
```

```
            for j in
```

```
range(len(hypothesis)):
```

```
                if hypothesis[j] !=
```

```
concepts[i][j]:
```

```
                    hypothesis[j] = '?'
```

```
print("Most Specific
```

```
Hypothesis:")
```

```
print(hypothesis)
```

OUTPUT:

```
.. Most Specific Hypothesis:
   ['?' '?' '?' '?']
```

2.

PROGRAM:

```
import pandas as pd

data = pd.DataFrame({
    'Study':['High','High','Low','Medium','High'],
    'Attendance':['Good','Good','Poor','Good','Average'],
    'Assignment':['Yes','Yes','No','Yes','Yes'],
    'Pass':['Yes','Yes','No','Yes','Yes']
})

concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values

hypothesis = None

for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis is None:
            hypothesis = concepts[i].copy()
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = '?'

print(hypothesis)
```

OUTPUT:

```
['?' '?' 'Yes']
```

3.

PROGRAM:

```
import pandas as pd

data = pd.DataFrame({
    'Income':['High','Medium','Low','High','Medium'],
    'Credit':['Good','Good','Poor','Excellent','Good'],
```

```

    'Job':['Yes','Yes','No','Yes','Yes'],
    'Loan':['Yes','Yes','No','Yes','Yes']
})

```

```

concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values

```

```

hypothesis = None

```

```

for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis is None:
            hypothesis = concepts[i].copy()
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = '?'

```

```

print(hypothesis)

```

OUTPUT:

```

['?' '?' 'Yes']

```

4.

PROGRAM:

```

import pandas as pd

```

```

data = pd.DataFrame({
    'Experience':['High','Medium','Low','High','Medium'],
    'Performance':['Good','Good','Poor','Excellent','Good'],
    'Training':['Yes','Yes','No','Yes','Yes'],
    'Promoted':['Yes','Yes','No','Yes','Yes']
})

```

```

concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values

```

```

hypothesis = None

```

```

for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis is None:
            hypothesis = concepts[i].copy()

```

```

else:
    for j in range(len(hypothesis)):
        if hypothesis[j] != concepts[i][j]:
            hypothesis[j] = '?'

print(hypothesis)

```

OUTPUT:

```
['?' '?' 'Yes']
```

5.

PROGRAM:

```

import pandas as pd

data = pd.DataFrame({
    'Age': ['Young', 'Young', 'Old', 'Middle', 'Young'],
    'Income': ['High', 'Medium', 'Low', 'Medium', 'High'],
    'Member': ['Yes', 'Yes', 'No', 'Yes', 'Yes'],
    'Buy': ['Yes', 'Yes', 'No', 'Yes', 'Yes']
})

concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values

hypothesis = None

for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis is None:
            hypothesis = concepts[i].copy()
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = '?'

print(hypothesis)

```

OUTPUT:

```
['?' '?' 'Yes']
```

RESULT :

Thus ,the FIND-S algorithm was implemented successfully ,and the most specific hypothesis was obtained .